Theory of Elementary Particles: Lecture 3

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Evaluation of some Feynman rules for QED

Elastic $e^- p \to e^- p$ cross section

Symmetries

P, T and C in classical physics

Lorentz group

Parity Invariance

$$
\begin{aligned}
* &= \overbrace{\psi(x)|\mathbf{p},r\rangle} && (1)\\
&= \int \frac{d^3\mathbf{k}}{\sqrt{2E_k}(2\pi)^3} \sum_s \left(a(\mathbf{k},s)u(\mathbf{k},s)e^{-\mathrm{i}\underline{k\cdot x}} + b^\dagger(\mathbf{k},s)v(\mathbf{k},s)e^{\mathrm{i}\underline{k\cdot x}} \right) \sqrt{2E_{\mathbf{p}}} a^\dagger(\mathbf{p},r)\,|0\rangle \\
&= \int \frac{d^3\mathbf{k}}{\sqrt{2E_k}(2\pi)^3} \sum_s \left(a(\mathbf{k},s)u(\mathbf{k},s)e^{-\mathrm{i}\underline{k\cdot x}} \right) \sqrt{2E_{\mathbf{p}}} a^\dagger(\mathbf{p},r)\,|0\rangle && (2)\\
&= u(\mathbf{p},r)e^{-\underline{ip\cdot x}}\,|\,0\rangle && (3)
\end{aligned}
$$

Analogously we compute the contraction with outgoing one-particle state

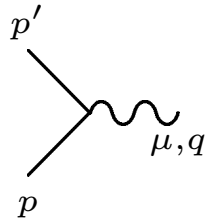
$$
\begin{aligned}
* &= \overbrace{\langle \mathbf{p},r\,|\,\overline{\psi}(x)} && (4)\\
&= \sum_s \langle 0\,|\,a(\mathbf{p},r)\sqrt{2E_{\mathbf{p}}} \int \frac{d^3\mathbf{k}}{\sqrt{2E_k}(2\pi)^3} \left(a^\dagger(\mathbf{k},s)\overline{u}(\mathbf{k},s)e^{\mathrm{i}\underline{k\cdot x}} + b(\mathbf{k},s)\overline{v}(\mathbf{k},s)e^{-\mathrm{i}\underline{k\cdot x}} \right) \\
&= \langle 0\,|\,\overline{u}(\mathbf{p},r)e^{\underline{ip\cdot x}} && (5)
\end{aligned}
$$

Now consider the antiparticle initial and final states,

$$
\begin{aligned}
* &= \overline{\psi}(x)|\mathbf{p}, r\rangle && (6)\\
&= \int \frac{d^3\mathbf{k}}{\sqrt{2E_k}(2\pi)^3} \sum_s \left(a^\dagger(\mathbf{k}, s)\overline{u}(\mathbf{k}, s)e^{\mathrm{i}\underline{k\cdot x}} + b(\mathbf{k}, s)\overline{v}(\mathbf{k}, s)e^{-\mathrm{i}\underline{k\cdot x}}\right)\sqrt{2E_{\mathbf{p}}}b^\dagger(\mathbf{p}, r)\,|0\rangle \\
&= \int \frac{d^3\mathbf{k}}{\sqrt{2E_k}(2\pi)^3} \sum_s b(\mathbf{k}, s)\overline{v}(\mathbf{k}, s)e^{-\mathrm{i}\underline{k\cdot x}}b^\dagger(\mathbf{p}, r)\,|0\rangle && (7)\\
&= \overline{v}(\mathbf{p}, r)e^{-\underline{ip\cdot x}}\mid 0\rangle && (8)
\end{aligned}
$$

for the outgoing state,

$$
\begin{aligned}
* &= \langle \mathbf{p}, r \mid \psi(x) && (9)\\
&= \langle 0 \mid b(\mathbf{p}, r)\sqrt{2E_{\mathbf{p}}} \int \frac{d^3\mathbf{k}}{\sqrt{2E_k}(2\pi)^3} \sum_s \left(a(\mathbf{k}, s)u(\mathbf{k}, s)e^{-\mathrm{i}\underline{k\cdot x}} + b^\dagger(\mathbf{k}, s)v(\mathbf{k}, s)e^{\mathrm{i}\underline{k\cdot x}}\right) \\
&= \langle 0 \mid v(\mathbf{p}, r)e^{\underline{ip\cdot x}} && (10)
\end{aligned}
$$



$$\sim \quad \langle \mathbf{p}', s \mid iT|\mathbf{p}, r; \underbrace{q, w}_{photon} \rangle \tag{11}$$

$$\tag{12}$$

$$= \quad \langle \mathbf{p}', s \mid T\left[-i\int d^4x \mathcal{H}_I(x)\right]|\mathbf{p}, r; \underbrace{q, w}_{photon} \rangle \tag{13}$$

$$= \quad \langle \mathbf{p}', s \mid T\left[-ie\int d^4x \bar{\psi}(x)\gamma^\mu\psi(x)A_\mu(x)\right]|\mathbf{p}, r; \underbrace{q, w}_{photon} \rangle \tag{14}$$

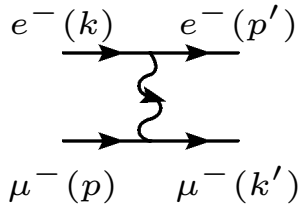
$$= \quad -ie\int d^4x \langle \mathbf{p}', s \mid T\left[\bar{\psi}(x)\gamma^\mu\psi(x)A_\mu(x)\right]|\mathbf{p}, r; q, w\rangle \tag{15}$$

$$= \quad -ie\int d^4x \langle \mathbf{p}', \overbrace{s \mid \bar{\psi}(x)}\gamma^\mu\overbrace{\psi(x)A_\mu(x)|\mathbf{p}, r}; q, w\rangle \tag{16}$$

$$= \quad \bar{u}(\mathbf{p}', s)(-ie\gamma^\mu)u(\mathbf{p}, r)\,\epsilon_\mu(w)\int d^4x\, e^{i(p'-p-q)\cdot x} \tag{17}$$

$$= \quad (2\pi)^4\delta^{(4)}(p'-p-q)\bar{u}(\mathbf{p}', s)(-ie\gamma^\mu)u(\mathbf{p}, r)\epsilon_\mu(w) \tag{18}$$

Interactions: QED examples



(19)

$$
\begin{aligned}
& * \quad = \\
& \frac{2\times(-ie)^2}{2}\int d^4x \int d^4y \langle \mathbf{p}', r'; \mathbf{k}', s' \mid \bar{\psi}(x)\gamma^\mu\psi(x)A_\mu(x)\bar{\psi}(y)\gamma^\nu\psi(y)A_\nu(y)|\mathbf{p}, r; \mathbf{k}, s; \rangle \\
& \qquad = \quad (-ie)^2\overline{u}(\mathbf{p}', r')\gamma^\mu u(\mathbf{k}, s)\ \overline{u}(\mathbf{k}', s')\gamma^\nu u(\mathbf{p}, r)\frac{-ig_{\mu\nu}}{(p'-k)^2+i\epsilon}
\end{aligned}
\tag{20}
$$

Diagrams drawn with use of feyn package but contraction made using simplewick

Elastic ep scattering

- Let us consider the process, in the lab frame,

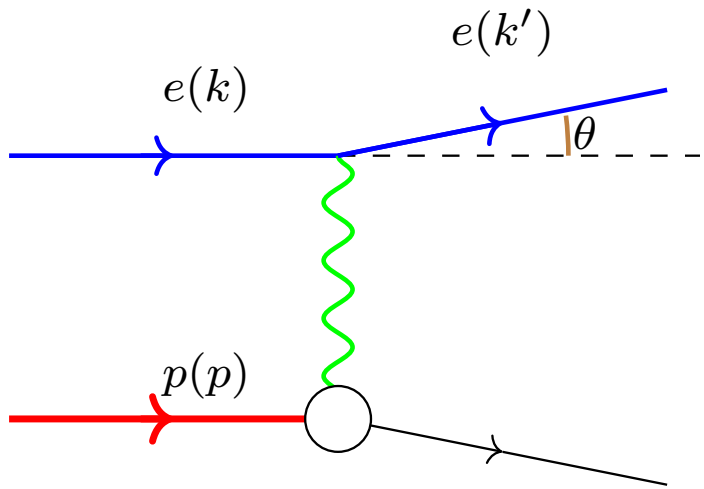
$$e^-(k) + p(p) \to e^-(k') + p(p')$$

- Let

$$k = (E, \mathbf{k}), \quad k' = (E', \mathbf{k}'), \quad \cos\theta = \frac{\mathbf{k}\cdot\mathbf{k}'}{|\mathbf{k}||\mathbf{k}'|}, \tag{21}$$

$$p = (M, 0), \quad p' = (E_{p'}, \mathbf{p}'), \tag{22}$$

where θ is the scattering angle.



- m and M denote electron and proton masses, respectively
- We define the four momentum transfer as:

$$q^\mu = p'^\mu - p^\mu = k^\mu - k'^\mu \tag{23}$$

- $$Q^2 = -q^2 = -2m^2 + 2k \cdot k' \tag{24}$$

  Notice that

$$-q^2 = -q_\mu q^\mu = -k^2 - k'^2 + 2k \cdot k' = -2m^2 + 2k \cdot k' \tag{25}$$

  Notice that:

$$q_0 = q^0 = E_{p'} - M = \sqrt{\mathbf{q}^2 + M^2} - M \geq 0 \tag{26}$$

$$Q^2 = -q^2 = -(p' - p)^2 = 2ME_{p'} - 2M^2 = 2M(E_{p'} - M) = 2q_0 M \geq 0 \tag{27}$$

$$d\sigma = \frac{1}{4EMv}\frac{d^3k'}{2E'(2\pi)^3}\frac{d^3p'}{2E_{p'}(2\pi)^3}(2\pi)^4\delta^{(4)}\left(k'+p'-p-k\right)|\mathcal{M}_{fi}|^2 \quad (28)$$

$$= \frac{1}{4EMv(2\pi)^2}\frac{d^3k'}{2E'}\frac{d^3p'}{2E_{p'}}\delta^{(4)}\left(k'+p'-p-k\right)|\mathcal{M}_{fi}|^2 \quad (29)$$

$$= \frac{1}{16EE'E_{p'}Mv(2\pi)^2}d^3k'd^3p'\delta^{(4)}\left(k'+p'-p-k\right)|\mathcal{M}_{fi}|^2 \quad (30)$$

$$= \frac{1}{16EE'E_{p'}Mv(2\pi)^2}d^3k'\delta\left(E'+E_{p'}-M-E\right)|\mathcal{M}_{fi}|^2 \quad (31)$$

where

$$v = \frac{|\mathbf{k}|}{E} \quad (32)$$

$$d\sigma = \frac{|\mathcal{M}_{fi}|^2}{16(2\pi)^2|\mathbf{k}|E'E_{p'}M} d^3k' \delta\left(E' + E_{p'} - E - M\right). \tag{33}$$

Notice that

$$d^3k' = |\mathbf{k}'|^2 d|\mathbf{k}'| d\Omega = 2\pi|\mathbf{k}'|E'dE'd\cos\theta \tag{34}$$

where

$$d\Omega = d\phi d\cos\theta$$

$$d\sigma = \frac{|\mathbf{k}'|\,|\mathcal{M}_{fi}|^2}{32\pi|\mathbf{k}|E_{p'}M} \delta\left(E' + E_{p'} - E - M\right) dE'd\cos\theta. \tag{35}$$

or

$$d\sigma = \frac{|\mathbf{k}'|^2\,|\mathcal{M}_{fi}|^2}{32\pi|\mathbf{k}|E'E_{p'}M} \delta\left(E' + E_{p'} - E - M\right) d|\mathbf{k}'|d\cos\theta. \tag{36}$$

Let us remind that

$$E'^2 = m^2 + \mathbf{k}'^2 \quad (37)$$

$$E^2_{p'} = \mathbf{q}^2 + M^2 \quad (38)$$

$$\mathbf{q}^2 = \mathbf{k}'^2 + \mathbf{k}^2 - 2|\mathbf{k}||\mathbf{k}'|\cos\theta \quad (39)$$

So

$$E^2_{p'} = \mathbf{k}'^2 + \mathbf{k}^2 - 2|\mathbf{k}||\mathbf{k}'|\cos\theta + M^2. \quad (40)$$

Then

$$* = \delta\left(\sqrt{\mathbf{k}'^2 + \mathbf{k}^2 - 2|\mathbf{k}||\mathbf{k}'|\cos\theta + M^2} + \sqrt{|\mathbf{k}'|^2 + m^2} - E - M\right) d|\mathbf{k}'| d\cos\theta \quad (41)$$

$$= \frac{\delta\left(|\mathbf{k}'| - k'\right)}{\left|\frac{|\mathbf{k}'|}{E'} + \frac{|\mathbf{k}'| - |\mathbf{k}|\cos\theta}{E_{p'}}\right|} \quad (42)$$

In the Lab frame $\mathbf{p}' = \mathbf{q}$

$$k' = \frac{(E+M)\sqrt{(E^2(M^2-m^2) + m^2(\mathbf{k}^2\cos^2\theta + m^2 - M^2))} + \sqrt{E^2-m^2}\cos\theta(EM+m^2)}{(E+M)^2 - |\mathbf{k}|^2\cos^2\theta} \quad (43)$$

(Let us note that the other root for k' is negative and can be rejected.)

Eventually the cross section reads

$$\frac{d\sigma}{d\cos\theta} = \frac{|\mathbf{k}'|^2\,|\mathcal{M}_{fi}|^2}{32\pi|\mathbf{k}|M|(E_{p'}+E')|\mathbf{k}'| - |\mathbf{k}|E'\cos\theta|} \tag{44}$$

$$= \frac{|\mathbf{k}'|^2\,|\mathcal{M}_{fi}|^2}{32\pi|\mathbf{k}|M|(E+M)|\mathbf{k}'| - |\mathbf{k}|E'\cos\theta|} \tag{45}$$

For $m \to 0$ we get

$$E' = k' = \frac{EM(E+M) + E^2 M\cos\theta}{(E+M)^2 - E^2\cos^2\theta} = \frac{EM}{E+M-E\cos\theta} \tag{46}$$

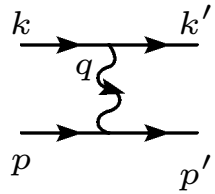
and the differential cross section:

$$\frac{d\sigma}{d\cos\theta} = \frac{E'^2\,|\mathcal{M}_{fi}|^2}{32\pi EE'M(E+M-E\cos\theta)} = \frac{E'^2}{32\pi E^2M^2}\,|\mathcal{M}|^2 \tag{47}$$

or written in a bit different form

$$\frac{d\sigma}{d\Omega} = \frac{E'^2}{64\pi^2E^2M^2}\left|\mathcal{M}_{fi}\right|^2, \tag{48}$$

$$e^-(k) + \mu^-(p) \to e^-(k') + \mu^-(p'), \quad q = k - k' \tag{49}$$



$$= \quad i\mathcal{M}_{fi} \tag{50}$$

$$= \quad \overline{u}(\mathbf{k}', s')(-ie\gamma^\mu)u(\mathbf{k}, s) \cdot \frac{-ig_{\mu\nu}}{q^2} \cdot \overline{u}(\mathbf{p}', r')(-ie\gamma^\nu)u(\mathbf{p}, r) \tag{51}$$

$$= \quad \frac{ie^2}{q^2} \cdot \overline{u}(\mathbf{k}', s')\gamma^\mu u(\mathbf{k}, s)\ \overline{u}(\mathbf{p}', r')\gamma_\mu u(\mathbf{p}, r) \tag{52}$$

Both vertices read $-ie\gamma$ since e^- and μ^- carry the same charge e. The beam and the target are given by a mixture of spin up and down components. So the initial wave function is given by

$$\psi(x) = \int \frac{d^3\mathbf{p}}{\sqrt{2E_p}(2\pi)^3} \sum_s \left(a(\mathbf{p}, s)u(\mathbf{p}, s)e^{-\mathrm{i}p\cdot x} + b^\dagger(\mathbf{p}, s)v(\mathbf{p}, s)e^{\mathrm{i}p\cdot x} \right), \tag{53}$$

The initial electron beam is a mixture of two spin states of electrons, the target has no spin specified, hence we must averaged over the two possible electron and two possible target spin states, so extra factor $1/4$, additionally we sum over all final spin states.

$$|\mathcal{M}_{fi}|^2 \to \frac{1}{4} \sum_{s,s',r,r'=\pm 1/2} |\mathcal{M}_{fi}|^2 = \star \tag{54}$$

$$\star = \frac{1}{4} \sum_{s,s',r,r'=\pm 1/2} |\mathcal{M}_{fi}|^2 \tag{55}$$

$$= \frac{e^4}{q^4}\frac{1}{4} \sum_{s,s',r,r'=\pm 1/2} \overline{u}(\mathbf{k}',s')\gamma^\mu u(\mathbf{k},s)\ \overline{u}(\mathbf{p}',r')\gamma_\mu u(\mathbf{p},r)$$

$$\times[\overline{u}(\mathbf{k}',s')\gamma^\nu u(\mathbf{k},s)\ \overline{u}(\mathbf{p}',r')\gamma_\nu u(\mathbf{p},r)]^* \tag{56}$$

$$= \frac{e^4}{q^4}\frac{1}{4} L^{\mu\nu} \times H_{\mu\nu} \tag{57}$$

Notice that

$$L^{\mu\nu} = \sum_{s,s'} \overline{u}(\mathbf{k}',s')\gamma^\mu u(\mathbf{k},s)[\overline{u}(\mathbf{k}',s')\gamma^\nu u(\mathbf{k},s)]^* \tag{58}$$

$$= \sum_{s,s'} \overline{u}(\mathbf{k}',s')\gamma^\mu u(\mathbf{k},s)\overline{u}(\mathbf{k},s)\gamma^\nu u(\mathbf{k}',s') \tag{59}$$

$$H_{\mu\nu} = \sum_{r,r'} \overline{u}(\mathbf{p}',r')\gamma_\mu u(\mathbf{p},r)[\overline{u}(\mathbf{p}',r')\gamma_\nu u(\mathbf{p},r)]^* \tag{60}$$

We used the identities

$$[\overline{u}(\mathbf{k}',s')\Gamma^{\nu}u(\mathbf{k},s)]^{*}=\overline{u}(\mathbf{k},s)\overline{\Gamma}^{\nu}u(\mathbf{k}',s'),\quad \overline{\Gamma}^{\nu}=\gamma_0\Gamma^{\nu\dagger}\gamma_0 \tag{61}$$

and

$$\sum_{s=\pm 1/2} u(\mathbf{k},s)\overline{u}(\mathbf{k},s)=\not{k}+m \tag{62}$$

Hence

$$\begin{aligned}
L^{\mu\nu} &= \mathrm{Tr}\gamma^{\mu}(\not{k}+m)\gamma^{\nu}(\not{k}'+m)=\mathrm{Tr}\gamma^{\mu}\not{k}\gamma^{\nu}\not{k}'+m^2\mathrm{Tr}\gamma^{\mu}\gamma^{\nu} && (63)\\
&= 4(k^{\mu}k'^{\nu}+k'^{\mu}k^{\nu}-g^{\mu\nu}(k\cdot k'-m^2)) && (64)\\
&\approx 4(k^{\mu}k'^{\nu}+k'^{\mu}k^{\nu}-g^{\mu\nu}k\cdot k') && (65)\\
H_{\mu\nu} &= 4(p_{\mu}p'_{\nu}+p'_{\mu}p_{\nu}-g_{\mu\nu}(p\cdot p'-M^2)) && (66)\\
&= 4(2p_{\mu}p_{\nu}+p_{\mu}q_{\nu}+q_{\mu}p_{\nu}-g_{\mu\nu}(p\cdot p'-M^2)) && (67)
\end{aligned}$$

So

$$\frac{d\sigma}{d\Omega}=\frac{E'^2e^4}{256q^4\pi^2E^2M^2}L_{\mu\nu}H^{\mu\nu}=\frac{E'^2\alpha^2}{16q^4E^2M^2}L_{\mu\nu}H^{\mu\nu} \tag{68}$$

$$e^2=4\pi\alpha$$

First notice that the electric currents are conserved

$$q_\mu \overline{u}(\mathbf{k}', s')\gamma^\mu u(\mathbf{k}, s) = 0 \quad (69)$$

$$q_\mu \overline{u}(\mathbf{p}', r')\gamma^\mu u(\mathbf{p}, r) = 0 \quad (70)$$

indeed

$$q_\mu \overline{u}(\mathbf{k}', s')\gamma^\mu u(\mathbf{k}, s) = \overline{u}(\mathbf{k}', s')(\not{k} - \not{k}')u(\mathbf{k}, s) = 0 \quad (71)$$

hence

$$q_\mu L^{\mu\nu} = q^\mu H_{\mu\nu} = 0 \quad (72)$$

$H_{\mu\nu}$ and $L_{\mu\nu}$ are symmetric.

Because the leptonic tensor is conserved, $q_\mu L^{\mu\nu} = 0$, the terms of $H_{\mu\nu}$ proportional to q_μ or q_ν do not contribute to the contraction $L^{\mu\nu} H_{\mu\nu}$:

$$\begin{aligned} H_{\mu\nu} &= 4(2p_\mu p_\nu + p_\mu q_\nu + q_\mu p_\nu - g_{\mu\nu}(p \cdot p' - M^2)) && (73) \\ &\rightarrow 4(2p_\mu p_\nu - g_{\mu\nu}(p \cdot p' - M^2)) && (74) \end{aligned}$$

Warning

This trick may be applied to **only one** of the two tensors: the reduced $H_{\mu\nu}$ no longer satisfies $q^\mu H_{\mu\nu} = 0$, hence an analogous reduction of $L^{\mu\nu}$ is not allowed anymore. We keep the **full** $L^{\mu\nu}$.

$$\begin{aligned} \frac{1}{16} L^{\mu\nu} H_{\mu\nu} &= \left(k^\mu k'^\nu + k'^\mu k^\nu - g^{\mu\nu}\, k \cdot k'\right) \left(2p_\mu p_\nu - g_{\mu\nu}(p \cdot p' - M^2)\right) \\ &= 4(k \cdot p)(k' \cdot p) - 2M^2(k \cdot k') + 2(k \cdot k')(p \cdot p' - M^2) && (75) \end{aligned}$$

We have assumed that $m \ll M$ and $m \approx 0$, hence

$$k\cdot k' = -\frac{q^2}{2}, \quad p\cdot p' = M^2 - \frac{q^2}{2}, \quad p\cdot k = \frac{s-M^2}{2}, \quad p\cdot k' = p\cdot k - p\cdot q = \frac{s-M^2+q^2}{2} \tag{76}$$

then

$$\frac{1}{16} L^{\mu\nu} H_{\mu\nu} = (s-M^2)(s-M^2+q^2) + M^2 q^2 + \frac{q^4}{2} \tag{77}$$

$$= \frac{1}{2}(s-M^2)^2 + \frac{1}{2}(s+q^2-M^2)^2 + M^2 q^2 \tag{78}$$

Notice that $s + q^2 - M^2 = -(u - M^2)$, so the result is explicitly symmetric under the $s \leftrightarrow u$ crossing.

Eventually the final formula reads

$$\frac{d\sigma}{d\Omega} = \frac{E'^2}{16\pi^2 E^2 M^2}\frac{e^4}{q^4}\left(\frac{1}{2}(s-M^2)^2 + \frac{1}{2}(s+q^2-M^2)^2 + M^2q^2\right) \tag{79}$$

Let us introduce the QED fine constant

$$\alpha = \frac{e^2}{4\pi} \approx \frac{1}{137} \quad e^2 = 4\pi\alpha \tag{80}$$

$$\frac{d\sigma}{d\Omega} = \frac{E'^2}{E^2 M^2}\frac{\alpha^2}{q^4}\left(\frac{1}{2}(s-M^2)^2 + \frac{1}{2}(s+q^2-M^2)^2 + M^2q^2\right) \tag{81}$$

Notice that in the Lab frame

$$s = (p+k)^2 \quad = \quad M^2 + 2ME \;\Rightarrow\; E = \frac{s-M^2}{2M} \tag{82}$$

$$q_0 \quad = \quad -\frac{q^2}{2M} \;\Rightarrow\; E' = E + \frac{q^2}{2M} = \frac{s+q^2-M^2}{2M} \tag{83}$$

then

$$\frac{E'}{E} = \frac{s+q^2-M^2}{s-M^2} = 1 + \frac{q^2}{s-M^2} < 1 \tag{84}$$

$$\frac{d\sigma}{d\Omega} = \frac{1}{M^2}\frac{\alpha^2}{q^4}\left(1+\frac{q^2}{s-M^2}\right)^2\left(\frac{1}{2}(s-M^2)^2 + \frac{1}{2}(s+q^2-M^2)^2 + M^2q^2\right) \tag{85}$$

What are the kinematic bounds for q^2 and s?

Symmetries

- Invariance of action:

$$S = \int d^4x \mathcal{L}(t, \mathbf{x}, \phi, ...) \rightarrow S = \int d^4x' \mathcal{L}(t', \mathbf{x}', \phi', ...) \tag{86}$$

- Invariance of density of Lagrangian

$$\mathcal{L}(t, \mathbf{x}, \phi, ...) \rightarrow \mathcal{L}(t', \mathbf{x}', \phi', ...) \tag{87}$$

- Invariance of density of Hamiltonian,

$$\mathcal{H}(t, \mathbf{x}, \phi, ...) \rightarrow \mathcal{H}(t', \mathbf{x}', \phi', ...) \tag{88}$$

- Invariance of $\mathcal{S}$-matrix, states and field operators...

$$\langle out \mid \mathcal{S} \mid in \rangle = e^{iphase} \langle out' \mid \mathcal{S}' \mid in' \rangle \tag{89}$$

$$= e^{iphase} \langle out \mid U^{-1}(sym)\, U(sym) \mathcal{S}\, U^{-1}(sym)\, U(sym) \mid in \rangle \tag{90}$$

So

$$\mathcal{O} \rightarrow U(sym) \mathcal{O}\, U^{-1}(sym) \tag{91}$$

- Exact and approximate symmetries
- Symmetry breaking

- Discrete symmetries $\mathcal{C}$, $\mathcal{P}$, $\mathcal{T}$ $\rightarrow$ Why?
- The results of experimental measurements $\rightarrow$ the fundamental discrete symmetries of particles and interactions
* Weak interactions breaks parity!(left-handed neutrinos)
- $\mathcal{CPT}$ theorem: fundamental statement of QFT

P, T and C in classical physics

$$\begin{aligned} \mathcal{P}: & \\ \mathbf{x} &\rightarrow -\mathbf{x} \\ \mathbf{p} &\rightarrow -\mathbf{p} \\ \mathbf{L} = \mathbf{x} \times \mathbf{p} &\rightarrow \mathbf{L} \end{aligned}$$

$$x^\mu \rightarrow x_\mu$$

$$\begin{aligned} \mathcal{T}: & \\ \mathbf{x} &\rightarrow \mathbf{x} \\ \mathbf{p} &\rightarrow -\mathbf{p} \\ \mathbf{L} &\rightarrow -\mathbf{L} \end{aligned}$$

$$x^\mu \rightarrow -x_\mu$$

$$\begin{aligned} \mathcal{T}: \quad x_\mu y^\mu &\rightarrow x_\mu y^\mu \\ \mathcal{P}: \quad x_\mu y^\mu &\rightarrow x_\mu y^\mu \end{aligned}$$

scalar in Minkowski space conserved

Charge invariance

- Reverse the charge of the particle $q \rightarrow -q$
- Reverse all generalized charges: Baryon, lepton numbers etc.

Lorentz group

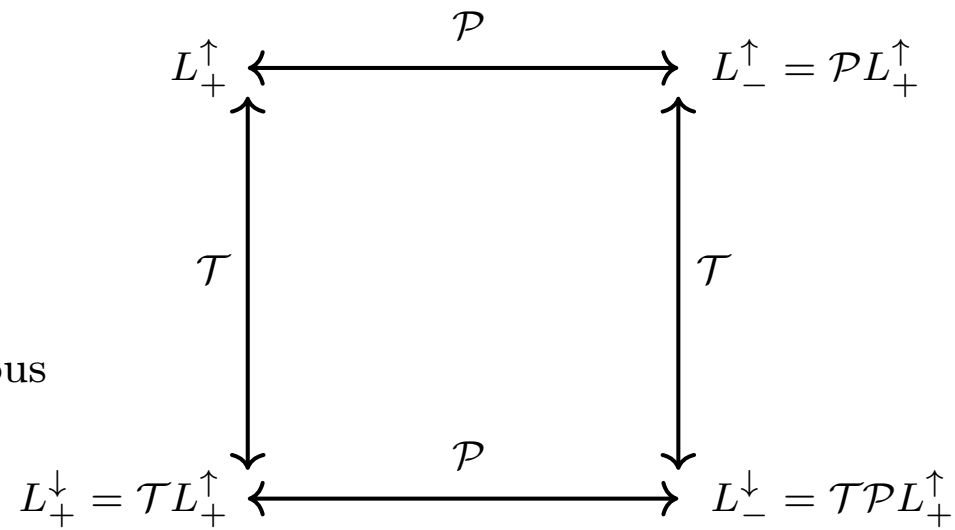
- Minkowski space: $x^\mu = (x^0, \mathbf{x})$, $x_\mu = g_{\mu\nu} x^\nu$
- Lorentz transformations keeps scalar product in Minkowski space invariant

$$
\begin{aligned}
x'^\mu &= \Lambda^\mu{}_\nu x^\nu, \quad \Lambda \in L && (92)\\
y'^\mu &= \Lambda^\mu{}_\nu y^\nu, && (93)\\
x'_\mu y'^\mu &= x_\mu y^\mu && (94)\\
x'_\mu y'^\mu &= g_{\nu\mu} \Lambda^\nu{}_\beta x^\beta \Lambda^\mu{}_\alpha y^\alpha && (95)\\
&= g_{\nu\mu} \Lambda^\nu{}_\beta \Lambda^\mu{}_\alpha x^\beta y^\alpha = g_{\beta\alpha} x^\beta y^\alpha && (96)\\
g_{\nu\beta} &= \Lambda^\mu{}_\nu g_{\mu\alpha} \Lambda^\alpha{}_\beta && (97)
\end{aligned}
$$

Lorentz group

$$
\begin{aligned}
x'^{\mu} &= \Lambda^{\mu}{}_{\nu} x^{\nu}, \quad \Lambda \in L \\
g_{\nu\beta} &= \Lambda^{\mu}{}_{\nu} g_{\mu\alpha} \Lambda^{\alpha}{}_{\beta}
\end{aligned}
$$

$$
\begin{aligned}
L^{\uparrow} &: \Lambda^{0}{}_{0} \geq 1 \quad \text{orthochronous} \\
L^{\downarrow} &: \Lambda^{0}{}_{0} \leq -1 \quad \text{non} - \text{orthochronous} \\
L_{+} &: \det\Lambda = +1 \quad \text{proper} \\
L_{-} &: \det\Lambda = -1 \quad \text{improper}
\end{aligned}
$$



- orthochronous: do not change the time arrow
- $L_{+}^{\uparrow}$: proper orthochronous

Lorentz invariance and field transformation

Let $x'^{\mu} = \Lambda^{\mu}{}_{\nu} x^{\nu}$

- Scalar field

$$\phi'(x') = \phi(x), \quad \text{i.e.} \quad U^{-1}(\Lambda)\phi(x')U(\Lambda) = \phi(\Lambda^{-1}x') \tag{98}$$

- Fermion field

$$\psi'(x') = S(\Lambda)\psi(x), \quad \text{i.e.} \quad U^{-1}(\Lambda)\psi(x')U(\Lambda) = S(\Lambda)\psi(\Lambda^{-1}x') \tag{99}$$

Parity is the space inversion. It can **not** be realized as a product of rotations ($\det = -1$), nevertheless it can be represented by a unitary transformation!

There exists linear operator P that performs space transformation on Hilbert space. It is chosen to be unitary.

$$\mathcal{P}^\dagger \mathcal{P} = 1 \quad \text{but} \quad \mathcal{P}^2 = 1 \tag{100}$$

$$\Rightarrow$$

$$\mathcal{P}^\dagger = \mathcal{P}^{-1} = \mathcal{P} \tag{101}$$

so $\mathcal{P}$ is hermitian and it is an observable.

- The eigenvalue of the state is called **parity** and it must be real so it reads $\eta_P = \pm 1$
- Parity is multiplicative quantum number!

If in the non-relativistic quantum mechanics parity and angular momentum is conserved

$$[\mathcal{P}, H] = 0 = [\mathcal{P}, \mathbf{L}] \tag{102}$$

The Hamiltonian, eigenstates have definite parity and angular momentum quantum number. Hence parity is a good quantum number to enumerate the states!

$$\mathcal{P}\mathbf{p}\mathcal{P}^{-1} = -\mathbf{p} \quad (103)$$

$$\mathcal{P} \mid p\rangle \rightarrow \mid -p\rangle \quad (104)$$

- A momentum eigenvector does not have a well-defined parity, and a parity eigenvector does not have a well-defined momentum!
- The state with $\mathbf{p} \neq 0$ can not be an eigenstate of the parity operator!

Property (exercise)

$$\mathcal{P}|l,m\rangle = (-1)^l |l,m\rangle,$$

where l is the orbital angular momentum

Parity and scalar field

$$LHS = \mathcal{P}\phi(t,\mathbf{x})\mathcal{P}^{-1} = \eta_B \phi(t,-\mathbf{x}) = RHS \tag{105}$$

B from boson and $\eta_B = \pm 1$
then

$$\begin{aligned} LHS &= \mathcal{P}\phi(t,\mathbf{x})\mathcal{P}^{-1} && (106)\\ &= \frac{1}{(2\pi)^3}\int \frac{d^3p}{\sqrt{2E_p}} \mathcal{P}\left[a(\mathbf{p})e^{-iE_\mathbf{p}t+i\mathbf{p}\cdot\mathbf{x}} + a^\dagger(\mathbf{p})e^{iE_\mathbf{p}t-i\mathbf{p}\cdot\mathbf{x}}\right]\mathcal{P}^{-1} && (107)\\ &= \frac{1}{(2\pi)^3}\int \frac{d^3p}{\sqrt{2E_p}} \left[\mathcal{P}a(\mathbf{p})\mathcal{P}^{-1}e^{-iE_\mathbf{p}t+i\mathbf{p}\cdot\mathbf{x}} + \mathcal{P}a^\dagger(\mathbf{p})\mathcal{P}^{-1}e^{iE_\mathbf{p}t-i\mathbf{p}\cdot\mathbf{x}}\right] && (108)\end{aligned}$$

on the other hand

$$\begin{aligned} RHS &= \eta_B\phi(t,-\mathbf{x}) && (109)\\ &= \eta_B\frac{1}{(2\pi)^3}\int \frac{d^3p}{\sqrt{2E_p}} \left[a(\mathbf{p})e^{-iE_\mathbf{p}t-i\mathbf{p}\cdot\mathbf{x}} + a^\dagger(\mathbf{p})e^{iE_\mathbf{p}t+i\mathbf{p}\cdot\mathbf{x}}\right] && (110)\\ &\quad \mathbf{p}\to -\mathbf{p} && (111)\\ &= \eta_B\frac{1}{(2\pi)^3}\int \frac{d^3p}{\sqrt{2E_p}} \left[a(-\mathbf{p})e^{-iE_\mathbf{p}t+i\mathbf{p}\cdot\mathbf{x}} + a^\dagger(-\mathbf{p})e^{iE_\mathbf{p}t-i\mathbf{p}\cdot\mathbf{x}}\right] && (112)\end{aligned}$$

Parity and scalar field

Hence

$$\mathcal{P} a^\dagger(\mathbf{p}) \mathcal{P}^{-1} = \eta_B \, a^\dagger(-\mathbf{p}) \tag{113}$$

$$\mathcal{P} a(\mathbf{p}) \mathcal{P}^{-1} = \eta_B \, a(-\mathbf{p}) \tag{114}$$

(η_B is real, $\eta_B = \pm 1$)
Let us assume that

$$\mathcal{P} \mid 0\rangle = + \mid 0\rangle \tag{115}$$

notice that

$$\mathcal{P} \mid \mathbf{p}\rangle = \sqrt{2E_p} \mathcal{P} a^\dagger(\mathbf{p}) \mid 0\rangle = \sqrt{2E_p} \mathcal{P} a^\dagger(\mathbf{p}) \mathcal{P}^{-1} \mathcal{P} \mid 0\rangle = \sqrt{2E_p} \eta_B \, a^\dagger(-\mathbf{p}) \mid 0\rangle = \eta_B \mid -\mathbf{p}\rangle \tag{116}$$

Only $\mathbf{p} = 0$ state is eigenstate of the parity operator

$$\mathcal{P} \mid \mathbf{p} = 0\rangle = \eta_B \mid \mathbf{p} = 0\rangle \tag{117}$$

with **intrinsic parity** $\eta_B = 1$ for scalar particle and -1 for pseudoscalar particle

Parity and Fermion field

By analogy coming from classical physics and the scalar field derivation we postulate:

$$\mathcal{P}a(\mathbf{p},s)\mathcal{P}^{-1} = \eta_a a(-\mathbf{p},s) \quad (118)$$
$$\mathcal{P}b(\mathbf{p},s)\mathcal{P}^{-1} = \eta_b b(-\mathbf{p},s) \quad (119)$$

Useful relation

$$\gamma_0 u(\mathbf{p},s) = u(-\mathbf{p},s) \quad (120)$$
$$\gamma_0 v(\mathbf{p},s) = -v(-\mathbf{p},s) \quad (121)$$

From the Dirac equation we have

$$\gamma_0(\gamma_0 p^0 - \boldsymbol{\gamma}\cdot\mathbf{p} - M)u(\mathbf{p},s) = (\gamma_0 p^0 + \boldsymbol{\gamma}\cdot\mathbf{p} - M)\gamma_0 u(\mathbf{p},s) = 0 \quad (122)$$
$$\Rightarrow \gamma_0 u(\mathbf{p},s) = u(-\mathbf{p},s) \quad (123)$$

The operator should transform as it follows

$$LHS = \mathcal{P}\Psi(t,\mathbf{x})\mathcal{P}^{-1} = S_P\Psi(t,-\mathbf{x}) = S_P\Psi(\tilde{x}) = RHS \tag{124}$$

$$\begin{aligned}
LHS &= \mathcal{P}\psi(t,\mathbf{x})\mathcal{P}^{-1} \qquad (125)\\
&= \mathcal{P}\frac{1}{(2\pi)^3}\int\frac{d^3p}{\sqrt{2E_p}}\sum_s\left\{a(\mathbf{p},s)u(\mathbf{p},s)e^{-ipx}+b^\dagger(\mathbf{p},s)v(\mathbf{p},s)e^{ipx}\right\}\mathcal{P}^{-1}\\
&= \frac{1}{(2\pi)^3}\int\frac{d^3p}{\sqrt{2E_p}}\sum_s\left\{\mathcal{P}a(\mathbf{p},s)\mathcal{P}^{-1}u(\mathbf{p},s)e^{-ipx}+\mathcal{P}b^\dagger(\mathbf{p},s)\mathcal{P}^{-1}v(\mathbf{p},s)e^{ipx}\right\}\\
&= \frac{1}{(2\pi)^3}\int\frac{d^3p}{\sqrt{2E_p}}\sum_s\left\{\eta_a a(-\mathbf{p},s)u(\mathbf{p},s)e^{-ipx}+\eta_b^* b^\dagger(-\mathbf{p},s)v(\mathbf{p},s)e^{ipx}\right\}\\
&\quad \mathbf{p}\to-\mathbf{p}, \quad \tilde{x}\equiv(t,-\mathbf{x})\\
&= \frac{1}{(2\pi)^3}\int\frac{d^3p}{\sqrt{2E_p}}\sum_s\left\{\eta_a a(\mathbf{p},s)u(-\mathbf{p},s)e^{-ip\tilde{x}}+\eta_b^* b^\dagger(\mathbf{p},s)v(-\mathbf{p},s)e^{ip\tilde{x}}\right\}\\
&= \frac{1}{(2\pi)^3}\gamma_0\int\frac{d^3p}{\sqrt{2E_p}}\sum_s\left\{\eta_a a(\mathbf{p},s)u(\mathbf{p},s)e^{-ip\tilde{x}}-\eta_b^* b^\dagger(\mathbf{p},s)v(\mathbf{p},s)e^{ip\tilde{x}}\right\} \qquad (126)
\end{aligned}$$

If one assumes that $\eta_a = -\eta_b^*$ then

$$\mathcal{P}\psi(t,\mathbf{x})\mathcal{P}^{-1} \quad = \quad \eta_a\gamma_0\psi(t,-\mathbf{x}), \quad \text{i.e.} \quad S_P = \eta_a\gamma_0 \tag{127}$$

Meson parity?

A meson is described by $q\overline{q}$ state, particle-antiparticle

$$\eta_a\eta_b = -|\eta_b|^2 = -1 = \eta_a^*\eta_b^* \tag{128}$$

then consider particle-antiparticle state and check its parity

$$\mathcal{P}a^\dagger(\mathbf{p},s)b^\dagger(\mathbf{p},s)|0\rangle = \mathcal{P}a^\dagger(\mathbf{p},s)\mathcal{P}^{-1}\mathcal{P}b^\dagger(\mathbf{p},s)\mathcal{P}^{-1}\mathcal{P}|0\rangle = -a^\dagger(-\mathbf{p},s)b^\dagger(-\mathbf{p},s)|0\rangle \tag{129}$$

if one assumes that vacuum is positive parity state, then particle-antiparticle state (for $\mathbf{p}=0$) is odd parity state!